

Karina Zueva

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EDUCATION

- 2024 – present **PGDip in Clinical Psychology** at Psychodemia, Russia
State-Certified Professional Retraining Program in Clinical Psychology.
- 2023 – 2025 **M.Sc. Psychology** at N.I. Lobachevsky National Research University, Nizhny Novgorod, Russia
Current average GPA is 5 (max. 5)
Major in Cyberpsychology, Human-Computer Interaction.
Selected courses: Statistical Data Analysis; Python Programming; Big Data; Cognitive Science; Psychophysiology; Neuroergonomics; Research Methods (Qualitative & Quantitative); Cyberpsychology.
- 2019 – 2023 **B.Sc. Psychology** at N.I. Lobachevsky National Research University, Nizhny Novgorod, Russia
Graduated with an average GPA of 4.98 (max. 5)
Major in Psychology.
Selected courses: Experimental Psychology; Mathematical Methods in Psychology; Data Processing; Technologies; Psychodiagnostics; Cognitive and Behavioral Psychology; Social and Organizational Psychology; Virtual Reality and Cyberpsychology.

ADDITIONAL EDUCATION

- 2026 – present **Code in Place, Stanford University, USA**
Introductory programming course in Python focused on software development fundamentals and problem solving.
- 2026 – present **Introduction to Machine Learning with Python, Arizona State University, USA**
Studying core machine learning concepts and Python-based tools.

RESEARCH ASSISTANT EXPERIENCE

- 2021 – 2025 **Driver Vigilance Monitoring System (Machine Vision, funded R&D project)**
Supervisor: Valeria Demareva, Ph.D.
N.I. Lobachevsky National Research University, Nizhny Novgorod, Russia
Cyberpsychology Laboratory
I contributed to the development of a driver vigilance monitoring system using machine vision techniques. My responsibilities included implementing components for detecting fatigue-related behavioral patterns and supporting the development of data-driven monitoring solutions.
- Video-Based Sleepiness Detection (Behavioral Annotation, funded R&D project)**
Supervisor: Valeria Demareva, Ph.D.
N.I. Lobachevsky National Research University, Nizhny Novgorod, Russia
Cyberpsychology Laboratory
I developed a comprehensive annotation framework for detecting sleepiness from video data, based on a systematic review of scientific literature. My work included identifying and formalizing behavioral markers of drowsiness (e.g., eye closure patterns, head movements), and creating a structured annotation scheme to enable consistent labeling across datasets. This framework was used to support subsequent video-based analysis and model development.
- Neuromarketing Eye-Tracking Study (external funded project)**
Supervisor: Valeria Demareva, Ph.D.
N.I. Lobachevsky National Research University, Nizhny Novgorod, Russia
Cyberpsychology Laboratory
I conducted experimental research using eye-tracking technology to analyze visual attention and user behavior. My responsibilities included working directly with participants, administering experimental sessions, and ensuring adherence to research protocols. I designed and implemented experimental procedures in PsychoPy, including stimulus presentation and behavioral data collection.
I also contributed to data preprocessing and the analysis of eye-tracking metrics to identify patterns of visual attention.

INDEPENDENT RESEARCH & APPLIED PROJECTS

- 2026 – present **Visual-Based Therapy (VBT): AI-Assisted Visualization Platform for Psychotherapy**
Ongoing project
Conceptualizing and developing a digital platform that externalizes internal therapeutic imagery into interactive, controllable visual environments; Conducted a structured review of imagery-based interventions (CBT, Schema Therapy, IFS, Compassion-Focused Therapy) and identified key limitations of current verbal-based approaches; Designed a system for creating personalized therapeutic scenarios with visual avatars, emotional tone modulation, and voice-guided interaction; Developing approaches for transforming internal cognitive processes (e.g., inner critic, emotional states, maladaptive beliefs) into interactive visual agents; Exploring integration of AI-generated avatars and dialogue systems to enhance therapeutic engagement and adherence; Investigating the role of visualization in increasing treatment effectiveness and user engagement
- 2025 – present **Human-AI Interaction and Loneliness: A Pilot Study**
Ongoing project
Supervisor: Valeria Demareva, Ph.D.
Conducted an experimental study investigating the effects of AI-based communication on perceived loneliness in young adults; Designed a 30-day longitudinal study with controlled interaction protocols and repeated measurements; Developed structured conversational scenarios (15 topics: abstract vs. personal) to analyze differences in emotional engagement; Collected and analyzed behavioral and self-report data from participants (N = 22) using validated psychological scales (UCLA Loneliness Scale, EETS, Big Five); Applied statistical methods including mixed-effects linear models (MixedLM), Mann-Whitney tests, and polynomial regression; Identified short-term reductions in loneliness following individual AI interactions, but no significant long-term effect; Demonstrated that emotionally relevant (personal) topics produce stronger effects than abstract discussions.

2023-2025	Virtual Reality Engagement and Perception Study <i>Supervisor: Roman Golubin, Ph.D.</i> I independently conducted a research project investigating user perception and engagement in virtual reality environments. Conducted a comprehensive literature review on immersion, presence, and engagement in virtual environments, synthesizing key theoretical frameworks; Designed and implemented the research study, including experimental procedures and behavioral data collection; Analyzed physiological data, including heart rate variability (HRV), using Python; Interpreted cognitive and behavioral responses in immersive environments, linking physiological indicators to user engagement; Contributed to the development of methodological approaches for measuring presence and immersion in VR.
2024	AI Avatar-Based Psychoeducation Course (CBT Introduction) Developed educational video materials introducing Cognitive Behavioral Therapy; Created scripts, generated AI-based video content, and structured learning materials; Designed digital learning experience and presentation interface.
2024	AI-Based Thought Tracking Application (CBT-oriented) Designed an application where AI analyzes recurring thought patterns and suggests CBT-based interventions; Conducted literature review on cognitive distortions and structured psychological models into app logic; Developed user journey, belief analysis system, and behavioral feedback mechanisms.
2023	Bot-Psychologist for Teenagers (CBT / DBT / ACT-based) Users: 500+ Developed a chatbot providing psychological support based on evidence-based approaches (CBT, DBT, ACT); Designed dialogue logic and emotional response system based on analysis of 50+ scientific sources; Implemented emotion classification (7 core emotional states) and decision-support scenarios.
2022	NERV: VR Creativity Enhancement System (Hackathon Project) Award: 3rd place on University Hackathon Designed a VR-based system to enhance individual and group creativity in organizational settings; Applied cognitive psychology frameworks (e.g., Six Thinking Hats, associative thinking techniques); Developed immersive scenarios to reduce distractions and facilitate idea generation.
2022	VR Game for Children's School Adaptation Concept development Award: 1st place, "Creative Industries: Learning to Manage VR Projects" competition Developed a VR-based intervention aimed at improving children's emotional intelligence, empathy, and school adaptation; Designed cognitive and social interaction scenarios based on psychological theory; Structured user journeys and behavioral tasks for assessing cognitive and social skills in immersive environments.

TEACHING AND CLINICAL EXPERIENCE

2025 – present	Lecturer in Consultative Psychology , N.I. Lobachevsky National Research University, online Developing and delivering lectures and practical classes in counseling psychology in Russian and English , designing assessments, and supervising students' academic work and thesis writing.
2021 – 2022	School Psychologist , Secondary School No. 62, Nizhny Novgorod, Russia Provided psychological counseling to students, parents, and teachers, conducted diagnostics and educational activities, and developed additional training programs.
2021 – 2023	Volunteer Psychologist (multiple settings) Delivered psychoeducational training for university students, provided mentoring for adolescents with disabilities, and assisted clinical psychologists in working with oncology patients.

GRANTS AND AWARDS

2019 - 2023	Award for Outstanding Academic and Research Achievements N.I. Lobachevsky National Research University, Nizhny Novgorod, Russia
2020	2nd Place, Interuniversity Psychology Olympiad , Kazan, Russia

CONFERENCES

2026	The Eighth International Forum on Cognitive Neuroscience "COGNITIVE NEUROSCIENCE SCIENCE – 2026", Ekaterinburg, Russia Report: The connection of communication with Artificial Intelligence and feelings of loneliness in young adults.
2023	Russian student Scientific conference with international participation "Issues of Psychophysiology – 2023", St. Petersburg, Russia Report: The impact of humanoid and non-humanoid avatars on feelings of presence in virtual environments.
2023	The Sixth International Forum on Cognitive Neuroscience "COGNITIVE NEUROSCIENCE SCIENCE – 2023", Ekaterinburg, Russia Report: Measuring the sense of presence in virtual environments through cardiac arrhythmia variability.
2022	International Scientific Conference "Topical aspects of modern psychophysiology", St. Petersburg, Russia Report: Experiencing the effect of presence in virtual reality games.

SKILLS

Research & Data Analysis:	Descriptive statistics, correlation analysis, t-tests, linear and logistic regression; Python (data analysis, Pandas, NumPy), Jupyter Notebook, SPSS
Data Acquisition:	EEG, eye-tracking, ECG, Virtual Reality (VR) equipment

**Clinical &
Psychological
Methods:
Languages**

CBT, REBT, clinical interviewing, case conceptualization, basic psychopathological assessment
Russian (native), English (B2, certified), German (A1).